Margeaux Corrigan

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EDUCATION

Brown University, *Sc. B. Electrical Engineering*, 4.0/4.0 GPA — Providence, RI | **Expected Graduation December 2029**

Relevant Coursework**:** Data Structures & Algorithms, Functional Programming, Statics & Dynamics, Ordinary Differential Equations

Fall 2026 Coursework: Computer Systems, Electricity & Magnetism, Probability & Statistics, Partial Differential Equations

TECHNICAL SKILLS

Programming Languages: C, Ada, Python, MATLAB, Java, OCaml, Racket

Embedded Systems: bare-metal firmware, register-level programming, I2C, SPI, UART, Arm Cortex-M0+ (RP2040), SPI flash logging

Hardware: Schematic design & PCB layout (KiCad), soldering, CAD (SolidWorks, Fusion 360), FEA (SolidWorks), 3D printing

Testing & Tools: Unity/CTest, Git/GitHub, CMake, GitHub Actions CI, OpenOCD, Alire, logic analyzer, oscilloscope, serial monitor

ENGINEERING PROJECTS

Bare-Metal Driver Stack, *Independent Project* — RP2040, C | May 2026 – Jul 2026

- Developed GPIO, SysTick, UART, I2C, and BMP390 drivers for RP2040 in C using register-level programming, no HAL
- Verified GPIO, UART, I2C, and BMP390 on hardware with a logic analyzer, OpenOCD, and serial monitor
- Built a Unity/CTest host test suite exercising every BMP390 driver function via link-time I2C and SysTick fakes
- Injected I2C NAK and timeout faults to validate error handling; automated cross-compilation and testing with GitHub Actions
- Traced a failed init to a BMP390 power-on value contradicting the datasheet; verified on hardware and documented

Rocket Payload Sensor System, *Sole Developer* — RP2040, Ada | Feb 2026 – Apr 2026

- Developed bare-metal Ada firmware for an RP2040 rocket payload under weekly review by an AdaCore engineer
- Deployed payload on a live high-power rocket launch; logged 377 m apogee, **16+ G** boost, and full descent to recovery
- Integrated BMP390 and LSM9DS1 sensors over I2C; captured boost, apogee, ejection, and descent in one record
- Built SPI flash logging pipeline for W25Q128; sized sample rate to fit a 21 h pre-launch window in 16 MB, logged 24.96 h
- Found a 1.5x count-to-g conversion error in post-flight analysis by checking logged data against expected static 1 G on the pad

EXPERIENCE

GE Aerospace, *Incoming Avionics Systems Engineering Co-op* — Fall 2027

- Selected for GE Aerospace LIFT Summit, 61 of 3,800+ applicants across 44 universities; co-op offer after interview

Interactive 3D Vision & Learning Lab (IVL), *UTRA Research Fellow* — Providence, RI | Jun 2026 – Present

Advisor: Srinath Sridhar, Assistant Professor | Brown University

- Supporting multi-camera rig development: PCB schematic & layout design, CAD modeling, & 3D-printed enclosure fabrication
- Assembled and functionally verified camera rig units; isolated failure causes and repaired units for return to service
- Ran data capture sessions and debugged false start/stop faults across the camera rig

Persistence Labs, *Strategy Intern* — Charleston, SC | May 2024 – Oct 2024

- Facilitated beta testing with high school test groups, collected feedback across 10+ iterations to identify usability issues
- Presented findings to founder/CEO and engineering team, directly informing feature prioritization and UI iteration

LEADERSHIP & ACTIVITIES

Brown Rocketry Club, *Avionics Team Lead, Incoming Co-President* — Providence, RI | Sep 2025 – Present

- Designed IREC avionics electrical architecture and wiring diagrams for power distribution, sensor interfaces, and flight systems
- Performed FEA on avionics bay components and developed CAM-based manufacturing plans in Fusion 360
- Secured industry sponsorship by leading partner outreach and presenting club technical work to external stakeholders

Brown Space Engineering, *Antenna Team Member* — Providence, RI | Mar 2026 – Present

- Designing communications antennas (Nitinol dipole, copper patch), including specification and integration constraints
- Establishing feed architecture and impedance matching strategy; documenting integration constraints on adjacent hardware

Project Amplify, *Co-Founder & Co-Director* — Irvine, CA | May 2024 – May 2025

- Co-founded a summer AI literacy program across 2 school districts; designed and taught curriculum, drove adoption efforts
- Taught students to train basic image recognition AI; introduced Python coding fundamentals and model evaluation

AWARDS & GRANTS

Brown UTRA Research Award, 2026 | **Brown Design Workshop Grant**, 2026 | **Taco Bell Ambition Accelerator Grant,** 2024